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Autism Insurance Mandate Generosity and Behavioral Treatment Use in Privately Insured Children with Autism Spectrum Disorder

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Master of Science in Public Health - 2026

Health Policy and Management

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Autism Insurance Mandate Generosity and Behavioral Treatment Use in Privately Insured Children with Autism Spectrum Disorder (ASD)

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Abstract

Autism Insurance Mandate Generosity and Behavioral Treatment Use in Privately Insured Children with Autism Spectrum Disorder (ASD)

By: Joseph Lee

Background: State autism insurance mandates vary in annual dollar caps and age eligibility, creating different coverage environments for privately insured children with autism spectrum disorder (ASD). Whether child-specific mandate generosity is associated with ASD-related behavioral treatment receipt remains unclear.
Methods: We pooled 2016–2023 National Survey of Children's Health data for 6,106 privately insured children ages 0–17 with caregiver-reported provider-diagnosed ASD. The primary outcome was caregiver-reported ASD behavioral treatment receipt in the past 12 months. Secondary outcomes were caregiver-reported problems paying the child's medical bills and unmet health care need in the past 12 months. The main independent variable was mandate generosity, operationalized as the child-specific annual dollar cap for autism behavioral treatment coverage, assigned based on state, survey year, and exact age and scaled in $10,000 units. We estimated survey-weighted logistic regression models with average marginal effects (AMEs), adjusting for predisposing, enabling, need-based, and contextual covariates, along with state and year fixed effects.
Results: In the past 12 months, 57.1% of caregivers reported that their child had received behavioral treatment, 28.9% reported problems paying medical bills, and 11.7% reported unmet healthcare need. In adjusted models, each $10,000 increase in mandate generosity was associated with a 1.23 percentage-point higher probability of ASD behavioral treatment receipt (95% CI, 0.31 to 2.16; p=0.009), and with a 0.85 percentage-point higher probability of unmet health care need (95% CI, 0.20 to 1.50; p=0.010). There was no significant association between mandate generosity and problems paying medical bills (AME, 0.65 percentage points; 95% CI, -0.19 to 1.50; p=0.131).
Conclusions: More generous autism behavioral treatment coverage caps were associated with higher ASD behavioral treatment receipt but also higher unmet healthcare need among privately insured children with ASD. Less restrictive mandate designs that expand age eligibility and increase or remove annual dollar caps may help improve access, but coverage generosity alone may be insufficient in improving treatment utilization without attention to workforce, plan design, and other access barriers.

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Contents

Chapter 1. Introduction and Significance

Introduction

Autism spectrum disorder (ASD) is a prevalent neurological and developmental condition that shapes how people communicate, interact with others, learn, and behave.[1] The most recent estimates from the CDC's surveillance network place ASD prevalence at approximately 1 in 31 U.S. children, highlighting the condition's substantial and growing reach as a public health concern.[2] As prevalence has risen, timely and equitable access to ASD-related services has become increasingly important for children and their families.[2]

Among available interventions, behavioral approaches (particularly applied behavior analysis [ABA]) occupy a central role in clinical guidelines; these interventions emphasize prompt identification and connection to developmental and behavioral services.[3-5] Evidence from early intervention studies suggests that structured behavioral therapies are associated with gains in language, communication, and adaptive functioning among young children with ASD.[4,6,7] Although some research suggests that greater treatment intensity may amplify these gains, findings on dose-response relationships remain inconsistent, and the optimal amount of intervention may vary across children.[8-10]

However, obtaining behavioral treatment can be burdensome for families. Intensive early ABA programs may involve up to 40 weekly hours of therapy and carry annual costs approaching $100,000.[11-13] Families seeking these services often face substantial financial burden and barriers to accessing specialty and therapy care, including unmet need and provider- or system-level access problems.[14-16] Narrow or inconsistent insurance coverage can compound these obstacles, contributing to unmet therapeutic need and difficulty sustaining regular treatment.[16] These barriers also are consequential from a health-systems perspective, given the substantial costs associated with childhood ASD and evidence that insurance coverage for ASD treatment shapes service use and spending.[17,18]

Policy Context

Starting in the early 2000s, U.S. states began adopting autism insurance mandates that required certain segments of the private insurance market to cover ASD-related services.[17,19,20] By 2019, every state and the District of Columbia had enacted at least one such mandate, though the scope and design of these statutes differ considerably.[12,13,20] Two design features are especially relevant: the age range within which a child qualifies for mandated coverage and the annual dollar ceiling placed on covered services.[12,21] These provisions jointly determine how much behavioral treatment a given child can access through private insurance in a particular state and year.[12,21]

Age eligibility thresholds can remove older children and adolescents from mandated benefits even when a state has an active mandate, while dollar caps (including those that vary by age tier) can restrict the total amount of treatment that insurers are required to cover within a year.[12,21-23] In contrast, some states impose neither age restrictions nor annual spending limits, producing comparatively generous coverage environments.[12,19,21] As a consequence, identifying only whether a state has a mandate in place can obscure substantial variation in the coverage that individual children actually experience. This variation is particularly consequential for behavioral treatment, where annual expenditures can accumulate quickly.[11]

Existing research indicates that adoption of autism mandates has been followed by increases in broader ASD-related health care use and spending among privately insured children, including outpatient ASD-specific services and behavioral and functional therapies.[17,25] Age caps, however, appear to dampen these gains as adolescents age out of eligibility.[21] Additional studies have found that state autism mandates were associated with increases in the ASD-related workforce, particularly among board-certified behavior analysts.[24] More recent descriptive policy studies have shifted from examining mandate adoption alone to comparing how states structure autism insurance mandates, particularly differences in age eligibility rules and annual spending caps, and using those features to characterize cross-state variation in mandate generosity.[12,13]

Problem Statement and Significance

Despite this growing evidence base, several gaps limit the current understanding of how mandate design relates to ASD-related behavioral treatment access. Many existing evaluations focus on broad outcomes such as treated prevalence, overall service use and spending, workforce growth, or the political determinants of mandate passage, rather than examining utilization of the ASD-specific behavioral treatments that mandates are primarily intended to support.[13,17,21,23-25] Furthermore, some studies draw on single-state or single-plan policy changes, yielding internally strong comparisons that may not extend to the full diversity of mandate structures found across the country.[25] Other investigations have documented cross-state variation in mandate benefit design, yet they stop short of constructing a child-level exposure that reflects the exact coverage available to a child at a specific age.[12,13,17,21] This gap is meaningful because age-tiered caps and coverage cutoffs can produce very different coverage realities for two children of different ages living in the same state and year.

The present study addresses these gaps by constructing a child-specific policy measure that reflects the annual dollar cap each child faces based on the intersection of state of residence, survey year, and exact age. Drawing on pooled 2016–2023 NSCH data, we test whether privately insured children with provider-diagnosed ASD who experience higher annual coverage limits are more likely to have received behavioral treatment in the past 12 months. We also examine whether this generosity measure is associated with two additional indicators of access and financial burden: caregiver-reported difficulty paying medical bills and unmet health care need. By constructing generosity at the level of the individual child rather than categorizing mandates as simply present or absent, this analysis offers a more granular and policy-relevant assessment of how specific design features may relate to differences in realized treatment access.

Research Question and Hypotheses

Research Question: Among privately insured children ages 0–17 with current provider-diagnosed ASD, is higher child-specific autism mandate generosity associated with a greater likelihood of receiving autism behavioral treatment in the past 12 months?

Secondary Research Question: Among the same population, is higher child-specific autism mandate generosity associated with a lower probability of problems paying medical bills or unmet health care need in the past 12 months?

Primary Hypothesis: Higher mandate generosity will be associated with a higher predicted probability of behavioral treatment receipt in the past 12 months after adjusting for predisposing, enabling, need-based, and contextual covariates and accounting for state and year fixed effects.

Secondary Hypotheses: Higher mandate generosity will be associated with (a) a lower probability of problems paying medical bills in the past 12 months and (b) a lower probability of unmet health care need in the past 12 months.

Conceptual Model

Exhibit 1 presents the study's conceptual model. The model draws on Andersen's Behavioral Model of Health Services Use, which organizes determinants of health care utilization into three broad categories: predisposing characteristics, enabling resources, and need-related factors, each operating at both individual and contextual levels.[26,27] This framework is a natural fit for the present study because it is designed specifically to explain utilization of health services. Its multilevel structure also accounts for the effect of individual and contextual level determinants of treatment receipt.[27]

Focal Relationship

At the center of the conceptual model is the hypothesized link between state mandate generosity and behavioral treatment utilization. The independent variable is a continuous, child-specific metric of mandate generosity that reflects the maximum annual dollar amount of ASD behavioral treatment coverage the child could access under the mandate rules operative in their state, year, and age. To construct this measure, a state-year-age policy file was built from primary statutory and regulatory text. Each child was then assigned an annual cap value according to three rules: (1) if the child's state had no mandate that year, or if the child's age fell outside the mandate's eligible range, the cap was set to $0; (2)

if the mandate included an annual dollar ceiling, the child received the age-specific cap applicable to them, including any tiered caps; and (3) if the mandate imposed no annual dollar limit, a placeholder of $100,000 was assigned, corresponding to the upper bound of published annual cost estimates for intensive early intervention.[13,28,29] The resulting values were scaled in $10,000 increments for interpretability. Because mandate eligibility and, in some states, annual dollar caps vary by child age, exact age is incorporated directly into the child-specific policy exposure.

The primary dependent variable is a dichotomous indicator, reported by the parent or caregiver, of whether the child received any behavioral treatment related to ASD in the past 12 months. Two secondary outcomes captured financial strain and access barriers: whether the family had problems paying medical bills and whether the child had any unmet health care need in the past 12 months.

We hypothesized that children experiencing higher effective caps would be more likely to receive behavioral treatment, with the primary mechanism being through expanded insurance coverage and lower financial barriers to accessing services. Additional hypothesized mechanisms include stronger reimbursement incentives that may encourage growth in provider supply and broader provider participation in covered autism services.[17,24,25]

Predisposing Characteristics

Predisposing factors are characteristics that precede current need and shape an individual's propensity to seek care, including demographic characteristics, social structure, and health beliefs.[26] The model includes child sex (male or female) and race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, and a combined Asian/Other category necessitated by small cell sizes). Prior research has documented racial and ethnic disparities in ASD service utilization, with children from minoritized backgrounds often showing lower participation in therapeutic or early intervention services than their non-Hispanic White peers.[15,30]

Enabling Resources

Enabling resources represent the material and organizational means through which families can act on a perceived need for care.[26] Individual-level enabling resources include parental education (high school or less; some college or associate degree; bachelor's degree or higher) and household income expressed as a percentage of the federal poverty level (below 100%, 100–199%, 200–399%, and 400% or above). Higher household income may help families absorb out-of-pocket costs associated with autism-related care, and greater parental education has been associated with greater use and intensity of intervention services.[31,32] Several additional enabling constructs are conceived but not directly measured in the analytic data. Health literacy, or the ability to access understand, appraise, and use health information and services, may shape how caregivers identify autism-related services and navigate referrals, insurance benefits, and authorization requirements; limited health literacy has been linked to greater difficulty with health system navigation and care-seeking.[33] Transportation access is another important enabling factor: families who lack reliable transportation face logistical and financial obstacles that can reduce service utilization, delay treatment initiation, or disrupt continuity of care.[34,35] Household structure may also influence treatment uptake; single-parent households may face greater caregiving and logistical demands, making coordination of health care services more difficult.[36,37]

Contextual-Level Enabling Characteristics

The focal relationship may also be shaped by area-level factors that vary across states and localities. Urbanicity is included because families in rural or non-metropolitan areas may face greater geographic barriers to ASD services and lower rates of health care use.[38,39] The models also include two state-level covariates to capture broader economic and workforce context: median household income (per $10,000) and psychiatrist supply per 100,000 persons. Although psychiatrist supply is an imperfect proxy for the ASD-specific behavioral workforce, it provides a general indicator of mental health service capacity.

Need-Based Factors

Need-based variables reflect the clinical characteristics that directly drive the demand for services.[26] The model includes parent-reported ASD severity (mild, moderate, or severe); counts of comorbid physical health conditions, counts of other comorbid neurodevelopmental conditions, and developmental delay, ADD/ADHD, anxiety, and depression. Children with greater ASD severity and co-occurring conditions tend to have higher service needs and greater health care use, which may in turn support greater treatment uptake.[40,41]

By synthesizing variation in mandate design with nationally representative data on children's treatment receipt, and by operationalizing generosity at the child level, this study offers a more refined assessment of how specific policy features may correspond to differences in behavioral treatment access. The findings may contribute to ongoing policy discussions about how to structure and improve coverage for privately insured children with ASD across the United States.

Chapter 2. Health Services Research Manuscript

Autism Insurance Mandate Generosity and Behavioral Treatment Use in Privately Insured Children with Autism Spectrum Disorder (ASD)

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Contribution Statement

This manuscript was developed in fulfillment of an MSPH thesis. Joseph Lee conceived the study question, compiled the state–year–age policy file, merged policy measures with the NSCH analytic dataset, conducted all statistical analyses, interpreted results, and drafted the manuscript and exhibits. Thesis committee members provided guidance and feedback on study design, analysis, interpretation, and presentation.

Abstract

Background: State autism insurance mandates vary in annual dollar caps and age eligibility, creating different coverage environments for privately insured children with autism spectrum disorder (ASD). Whether child-specific mandate generosity is associated with ASD-related behavioral treatment receipt remains unclear.

Methods: We pooled 2016–2023 National Survey of Children's Health data for 6,106 privately insured children ages 0–17 with caregiver-reported provider-diagnosed ASD. The primary outcome was care-giver reported ASD behavioral treatment receipt in the past 12 months. Secondary outcomes were dichotomous outcomes were caregiver-reported problems paying the child's medical bills and unmet health care need. The main independent variable was mandate generosity, operationalized as the child-specific annual dollar cap for autism behavioral treatment coverage, assigned based on state, survey year, and exact age and scaled in $10,000 units. We estimated survey-weighted logistic regression models with average marginal effects (AMEs), adjusting for predisposing, enabling, need-based, and contextual covariates, along with state and year fixed effects.

Results: In the past 12 months, 57.1% of caregivers reported that their child had received behavioral treatment, 28.9% reported problems paying medical bills, and 11.7% reported unmet healthcare need. In adjusted models, each $10,000 increase in mandate generosity was associated with a 1.23 percentage-point higher probability of ASD behavioral treatment receipt (95% CI, 0.31 to 2.16; p=0.009), and with a 0.85 percentage-point higher probability of unmet health care need (95% CI, 0.20 to 1.50; p=0.010). There was no significant association between mandate generosity and problems paying medical bills (AME, 0.65 percentage points; 95% CI, -0.19 to 1.50; p=0.131).

Conclusions: More generous autism behavioral treatment coverage caps were associated with higher ASD behavioral treatment receipt but also higher unmet healthcare need among privately insured children with ASD. Less restrictive mandate designs that expand age eligibility and increase or remove annual dollar caps may help improve access, but coverage generosity alone may be insufficient in improving treatment utilization without attention to workforce, plan design, and other access barriers.

Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental condition that influences communication, social interaction, learning, and behavior.[1] Recent surveillance estimates from the Centers for Disease Control and Prevention indicate that ASD is now identified 1 in 31 children in the United States, highlighting the condition's substantial and growing reach as a public health concern.[2] Children with ASD often benefit from early identification and timely referral to appropriate behavioral and developmental services.[3,4] Behavioral interventions, particularly applied behavior analysis (ABA), are widely considered the gold-standard of evidence-based care for children with ASD, and clinical guidance recommends timely identification and referral to appropriate behavioral and developmental services.[3-5,42] Early use of these treatment models is consistently associated with improved developmental outcomes and prognosis, including gains in communication, language, and adaptive functioning.[4,6,7] Although some research suggests that greater treatment intensity may amplify these gains, findings on dose-response relationships remain mixed, and the optimal amount of intervention may vary across children.[8-10]

However, obtaining behavioral treatment can be burdensome for families. Early ABA programs may require up to 40 hours per week and can cost as much as $100,000 annually.[11-13] Families seeking these services often face substantial financial burden and barriers to accessing specialty and therapy care, including unmet need and provider- or system-level access problems.[14-16] Narrow or inconsistent insurance coverage can compound these obstacles, adding to unmet therapeutic need and difficulty sustaining regular treatment.[16]

Beginning in the early 2000s, states have enacted autism insurance mandates to expand coverage of ASD-related services within segments of the private insurance market.[17,19,20] However, these mandates vary substantially across states in two design features most relevant to this study: age eligibility rules and annual dollar caps.[12,21] Together, these provisions determine the effective amount of behavioral treatment coverage available to a child under a mandate in a given state-year.[12,21] Age-based eligibility rules can exclude older children and adolescents even in mandate states, while dollar caps—including tiered caps that vary by age—can limit how much behavioral treatment insurance must cover over a year.[12,21-23] These design features are especially important because service use patterns vary across childhood, with use of therapeutic services generally declining with age among children with ASD.[22] By contrast, some states require coverage with no age restrictions and no annual dollar caps whatsoever, creating more generous coverage environments.[12,19,21] As a result, mandate status alone can mask meaningful heterogeneity in the coverage available to children. These differences are especially consequential for autism behavioral treatment, where spending can accumulate rapidly.[6]

Prior studies suggest that autism insurance mandates can increase broader ASD-related health care utilization among privately insured children, although this association appears to be attenuated by age caps as children age out of coverage.[17,21,25] For example, a national claims-based analysis by Barry and colleagues found that state mandates were associated with increased service use and spending for privately insured children with ASD, including outpatient ASD-specific services and behavioral and functional therapies.[17] Additional studies have found that state autism mandates were associated with increases in the ASD-related workforce, particularly among board-certified behavior analysts.[24] More recent descriptive policy studies have shifted from examining mandate adoption alone to comparing how states structure autism insurance mandates, particularly differences in age eligibility rules and annual spending caps, and using those features to characterize cross-state variation in relative mandate generosity.[12,13]

Despite this growing evidence base, several gaps limit the current understanding of how mandate design relates to ASD-related behavioral treatment access. Much of the literature on mandates focuses on treated prevalence, overall service use or spending, workforce effects, or mandate adoption, rather than treatment-specific utilization that is most directly targeted by these policies.[13,17,21,23-25] In addition, some evaluations rely on single-state or single-plan policy changes, which may not generalize to the full range of mandate designs nationwide.[25] Other studies document the cross-state variation in mandate benefit parameters—including age eligibility rules and spending caps—but do not link that policy variation to child-level differences in treatment uptake.[12,13,17,21] This limitation is particularly important because tiered caps and age-based cutoffs can create qualitatively different policy environments: two children in mandate states may face very different coverage limits depending on their exact age, even within the same state-year.[12]

The present study addresses these gaps by linking detailed, age-specific policy variation to child-level outcomes in a nationally representative sample. Using pooled 2016–2023 NSCH data, we examine whether privately insured children with provider-diagnosed ASD who are exposed to more generous mandate provisions— measured as the child's specific annual behavioral treatment coverage cap given state, survey year, and exact age —are more likely to receive autism behavioral treatment. We also assess whether mandate generosity is associated with two additional policy-relevant measures of access: caregiver-reported problems paying medical bills and unmet health care need. By measuring generosity at the child level rather than treating mandates as a uniform, yes-or-no exposure, this study provides a policy-relevant test of how mandate design features may translate into differences in realized access.

Methods

Data

We used data from the National Survey of Children's Health (NSCH) public-use files from 2016 through 2023. The NSCH is a nationally representative survey of noninstitutionalized U.S. children ages 0–17, completed by a parent or caregiver, and it is designed for national and state-level inference when survey weights and design variables are applied.[26] Policy-level data were compiled by the author from primary state statutory and regulatory text. Mandate parameters, including effective dates, age eligibility thresholds, and annual dollar caps, were abstracted from the legal text and harmonized into a state-year-age policy file. Consistent with prior state autism mandate generosity studies, key mandate features were cross-referenced against publicly available secondary policy summaries—including the National Conference of State Legislatures and Autism Speaks—to verify completeness through 2023, and discrepancies were resolved using the primary policy text.[12,13,43,44]

Sample Derivation

The analytic sample was drawn from pooled 2016–2023 National Survey of Children's Health cycles and was restricted to children ages 0–17 with private health insurance and a history of provider-diagnosed ASD. Applying these inclusion criteria yielded a base sample of 6,463 children. From this group, we excluded children with missing data on the primary outcome of ASD behavioral treatment in the past 12 months (n=29), missing data on secondary access outcomes (n=10), and missing demographic, socioeconomic, need-based, contextual, or comorbidity covariates (n=318). The final analytic sample included 6,106 children. Additional exclusion detail is presented in Exhibit 2. Because the problems paying medical bills item is asked conditionally in the NSCH and missing values on this outcome were excluded from the regression model, the model-specific sample for that outcome-specific model was smaller (N=5,435).

Measures

Outcomes

The primary outcome was parent-reported receipt of ASD behavioral treatment during the past 12 months. In the NSCH, behavioral treatment is a broad category capturing whether the child or parent received training or an intervention to help with behaviors related to ASD. This outcome should be interpreted as receipt of any ASD-related behavioral intervention within the past 12 months. Additional detail on the scope and limitations of this measure is discussed in the limitations section. Secondary outcomes included two binary measures capturing financial hardship and access barriers in the prior year: whether the family had problems paying medical bills during the past 12 months and whether the child had any unmet health care need during the past 12 months.

Policy exposure

The primary policy exposure was a continuous, age-specific measure of autism insurance mandate generosity for ASD-related behavioral therapy services that captures the maximum annual dollar amount of ASD-related coverage available to a child given their state of residence, survey year, and exact age.

Using the state–year policy file, each state-year mandate was assigned an age-specific annual cap. If a state had no autism mandate in that year or the child's age fell outside the mandate's eligible age range, the effective cap was set to $0. For mandates with annual dollar caps, children whose age fell within the eligible range were assigned the exact age-specific annual cap, including tiered caps that vary by age where applicable. For mandates with no annual dollar maximum, the effective cap was set to a fixed placeholder value of $100,000. The $100,000 placeholder was set to the upper end of published estimates for the annual cost of intensive early intervention for children with ASD, to operationalize "no annual dollar maximum" mandates as substantially more generous than age/dollar-capped mandates.[11-13] This state–year–age policy panel was then merged to the NSCH analytic sample by state (FIPS), survey year, and child age to assign each child a single exposure value. For interpretability in regression models, the final exposure was scaled in $10,000 units, so a one-unit increase corresponds to a $10,000 higher effective annual coverage cap available to that child in their state and survey year (with $0 indicating no mandate coverage or age ineligibility).

Conceptual Framework

Covariate selection and the study's hypothesized relationships were guided by Andersen's Behavioral Model of Health Services Use, which conceptualizes health care utilization as a function of predisposing characteristics, enabling resources, and need-related factors at both the individual and contextual levels.[26,27] Exhibit 1 presents the study's conceptual model. We hypothesized that greater mandate generosity would be positively associated with autism behavioral treatment receipt because more generous mandate designs may expand ASD service coverage, reduce financial barriers, and support provider participation and provider supply, thereby improving access to behavioral treatment among privately insured children with ASD.[17,21,24,45]

Covariates

Predisposing characteristics included child sex (male [reference], female) and race/ethnicity (non-Hispanic White [reference], non-Hispanic Black, Hispanic, and Asian/Other). Enabling resources included parental education (high school or less [reference], some college or associate degree, and bachelor's degree or higher) and household federal poverty level (FPL) category (less than 100% FPL [reference], 100–199% FPL, 200–399% FPL, and 400% FPL or above). Contextual enabling characteristics included urbanicity (non-metropolitan [reference], metropolitan, and unknown/missing). Need-related factors included parent-reported ASD severity (mild [reference], moderate, severe, and unknown/missing); counts of comorbid physical conditions (0 [reference], 1, or 2+; see Appendix Table A1 for conditions included); counts of comorbid neurologic conditions (0 [reference], 1, or 2+; see Appendix Table A1 for conditions included); and dichotomous indicators for developmental delay, attention deficit disorder/attention deficit hyperactivity disorder (ADD/ADHD), anxiety, and depression. For ASD severity and urbanicity, we retained observations by including an explicit missing category where applicable.

Analysis

We first summarized weighted sample characteristics and weighted ASD behavioral treatment prevalence across covariate strata (Exhibit 3) and examined unadjusted bivariate associations between the continuous policy exposure and each outcome using survey-adjusted linear regression with pairwise design-based F tests (Exhibit 4). We then estimated survey-weighted logistic regression models for each outcome. In unadjusted models, the outcome was modeled as a function of the policy exposure only. Adjusted models included the full covariate set plus state and survey-year fixed effects, yielding a two-way fixed effects specification that leverages within-state changes in mandate generosity over time while accounting for time-invariant state characteristics and national secular trends shared across states. The exposure was entered into the primary model in $10,000 units, and we report average marginal effects (AMEs), representing the average absolute change in predicted probability associated with a one-unit increase in the exposure. All analyses accounted for the NSCH complex sampling design using survey weights and design variables (strata and primary sampling units) to produce nationally representative estimates for the analytic sample. Analyses were conducted in Stata (version 17).

Results

The analytic cohort included 6,106 privately insured children ages 0–17 with provider-diagnosed ASD (Exhibit 2). The sample was predominantly male (77.9%) and non-Hispanic White (56.7%), with the majority of families reporting household incomes at or above 200% of the federal poverty level (75.2%) and parental education of a bachelor's degree or higher (65.8%). Nearly half of the sample had moderate or severe ASD (43.9%). Moreover, comorbid conditions were common, with 66.6% having developmental delay and 43.5% having ADD/ADHD. Weighted descriptive characteristics and weighted behavioral treatment prevalence across covariate strata are shown in Exhibit 3.

In bivariate analyses, policy generosity was not associated with ASD behavioral treatment receipt ($p=0.205$), problems paying medical bills ($p=0.070$), or unmet health care need ($p=0.956$) (Exhibit 4).

In adjusted models with year and state fixed effects (Exhibit 6), each $10,000 increase in mandate generosity was associated with a 1.23 percentage-point higher probability of ASD behavioral treatment receipt (95% CI, 0.31 to 2.16; $p=0.009$). Each $10,000 increase in mandate generosity was also associated with a 0.85 percentage-point higher probability of unmet health care need (95% CI, 0.20 to 1.50; $p=0.010$). Associations with problems paying medical bills were not statistically significant (AME, +0.65 percentage points; 95% CI, -0.19 to 1.50; $p=0.131$).

ASD behavioral treatment receipt was also strongly associated with several covariates. Compared with children reported to have mild ASD, children with severe ASD had a 32.44 percentage-point higher predicted probability of behavioral treatment receipt ($p<0.001$), and those with moderate ASD had a 14.89 percentage-point higher predicted probability ($p<0.001$). Developmental delay ($p<0.001$) and anxiety ($p=0.008$) were also associated with higher treatment receipt. Treatment receipt was higher among children living in metropolitan areas ($p=0.003$) and among those in the highest income category (≥400% FPL; $p=0.011$).

Problems paying medical bills and unmet health care need were primarily associated with socioeconomic status and health complexity. Higher household income (≥400% FPL) was associated with a lower probability of problems paying medical bills ($p<0.001$) and a lower probability of unmet need ($p=0.038$). Compared with children with no physical comorbidities, children with one physical condition (problems paying: $p=0.017$; unmet need: $p=0.045$) and those with two or more physical conditions (problems paying: $p<0.001$; unmet need: $p=0.001$) had higher probabilities of both outcomes. Anxiety, depression, and ADD/ADHD were each associated with higher unmet need (all $p<0.05$), and anxiety and ADD/ADHD were associated with higher problems paying medical bills ($p=0.002$ and $p=0.035$, respectively).

Sensitivity analyses using alternate categorical encodings of policy exposure showed a similar overall pattern. In the 3-category specification, exposure to an effective cap of ≥$36,000 was associated with a higher probability of ASD behavioral treatment receipt (AME, +15.34 percentage points; 95% CI, 6.46 to 24.22; $p=0.001$) and unmet health care need (AME, +6.64 percentage points; 95% CI, 2.56 to 10.70; $p=0.001$), whereas the intermediate category (>$0 to <$36,000) was not significantly associated with any outcome. In the binary specification, any positive effective cap was associated with a higher probability of ASD behavioral treatment receipt (AME, +11.78 percentage points; 95% CI, 3.34 to 20.22; $p=0.006$) and unmet health care need (AME, +4.63 percentage points; 95% CI, 0.81 to 8.45; $p=0.018$). Associations with problems paying medical bills were not statistically significant in either sensitivity analysis.

Discussion

Using a child-specific measure of autism insurance mandate generosity that incorporates both age eligibility and annual dollar caps, this study examined behavioral treatment and two broader access outcomes among privately insured children with ASD. We found three main results. First, more generous mandate exposure was associated with a higher probability of caregiver-reported behavioral treatment receipt. Second, more generous exposure was also associated with a higher probability of unmet health care need. Third, mandate generosity was not significantly associated with problems paying medical bills.

The positive association with behavioral treatment receipt is broadly consistent with prior evaluations showing that state autism insurance mandates were followed by increases in ASD-related service use and spending in commercially insured populations.[17,25] A key implication is that more generous, less restrictive mandate design features may be associated with increased utilization. Age eligibility limits can create discontinuities in mandated coverage as children grow older, and annual dollar caps can constrain access when families seek intensive behavioral services.[21]

Interpretation of the policy coefficient also requires care because the exposure intentionally combines two dimensions of policy: moving from $0 to a positive effective cap reflects both mandate presence and age-based eligibility, whereas increases among children already above $0 reflect greater generosity of the annual maximum, including shifts from capped to no-cap designs. Although the average marginal effect appears modest on a per-$10,000 basis (AME = 1.23 percentage points), it accumulates meaningfully across the policy range observed in the study period. This accumulation is substantively meaningful because autism mandate design has evolved toward broader, less restrictive coverage in many states, particularly with respect to age eligibility and benefit limits—in our policy panel, we likewise observed no downward movement in generosity during the study period.[12] Adjusted predicted probabilities indicated a clear upward gradient with mandate generosity; as shown in Exhibit 8, moving from no mandated coverage to a no-cap design corresponds to an approximately 12 percentage-point higher predicted probability of behavioral treatment receipt.

The positive association between higher effective caps and unmet health care need is plausible for several reasons. National survey evidence indicates that children with ASD experience substantial unmet need, particularly for specialty and therapy services, consistent with persistent access barriers.[16] In addition, the NSCH unmet-need item is not autism-specific; it captures whether the child did not receive needed health care in the past 12 months, regardless of condition. Children with ASD often have higher comorbidity burden, substantial medical and behavioral service needs, and higher health care use overall.[18,46] Thus, if generous ASD coverage increases engagement with the health system, families may also recognize or report unmet needs in other domains (specialty care, mental and physical health, etc.). More generous caps may also increase care-seeking and utilization; if demand rises faster than provider availability, unmet need can increase even when coverage is more generous on paper.[17,24]

The lack of a statistically significant association between mandate generosity and problems paying medical bills likely reflects the breadth of that outcome. It captures overall health care financial strain rather than ASD-specific spending, and autism behavioral treatment is only one component of a child's broader health care use. Even if coverage for ASD behavioral treatment becomes more generous, that may not translate into measurable reductions in overall family financial strain. Prior mandate evaluations likewise have not consistently found reductions in family financial burden.[47]

From a policy perspective, these findings suggest that increasing the generosity of mandate design—particularly by extending age eligibility and increasing or removing annual dollar maximums—may improve access to autism behavioral treatment among privately insured children with ASD. However, the positive association with unmet health care need indicates that mandate design may be only one determinant of access and families' healthcare experience; there may be a need for complementary policies that address provider supply, network adequacy, (for example, adequate reimbursement and administrative processes) and system-level affordability so that expanded coverage truly translates into accessible care for families.

Limitations

This study has several limitations. First, the NSCH is cross-sectional and relies on caregiver report. Second, estimates may be subject to residual confounding from unmeasured time-varying state factors correlated with mandate generosity, such as changes in private insurance regulation, market composition, reimbursement, network design, or provider supply. Third, the primary outcome is a dichotomous measure of any ASD behavioral treatment in the prior 12 months and does not distinguish modality, intensity, duration, or continuity; children may also receive behavioral services through school or early intervention systems that do not flow through private insurance. Fourth, the policy exposure primarily reflects annual dollar caps attached to autism behavioral treatment benefits—often written around ABA in state mandate text—whereas the NSCH outcome captures a broader parent-reported ASD-related behavioral treatment category. This mismatch may attenuate associations. Finally, state mandates do not apply uniformly across all private plans, particularly self-insured plans governed by ERISA; thus, incomplete exposure to mandates among privately insured children may attenuate estimated associations.[43]

Conclusion

Using nationally representative NSCH data from 2016 to 2023, higher autism mandate generosity was associated with a higher probability of caregiver-reported ASD behavioral treatment receipt among privately insured children with ASD. More generous mandates were also associated with a higher probability of unmet health care need, while associations with problems paying medical bills were not statistically significant. Together, these findings suggest that more generous mandate design features may be associated with increased behavioral treatment utilization, but coverage generosity alone may be insufficient to improve treatment utilization without attention to workforce, plan design, and other access barriers.

Future research should examine the mechanisms through which mandate generosity may operate—such as whether increased spending or broader coverage translates into greater utilization—as well as the barriers that persist even in more generous mandate environments, including provider availability, wait times, network adequacy, prior authorization, and out-of-pocket costs. Integrating contextual state measures and plan-level features could help clarify the roles of provider supply, network participation, and reimbursement practices. Claims-based or EHR data could distinguish ABA from other behavioral interventions and provide more contextualized measures of treatment utilization, including intensity, frequency, modality, and continuity. This additional granularity of measurement would allow researchers to assess how more generous mandates translate into service utilization, spending, and family financial burden. Qualitative, family-centered research could also help further clarify the challenges families face in accessing care and the types of care and support they find most valuable.

Chapter 3. Expanded Discussion

What This Study Adds

This analysis linked a child-specific measure of autism mandate generosity to caregiver-reported behavioral treatment receipt among privately insured children with ASD using pooled 2016–2023 NSCH data. The exposure variable captures the maximum annual dollar amount of ASD-specific behavioral treatment coverage available to each child, determined by the combination of the child's state of residence, survey year, and exact age. This approach moves beyond prior studies by accounting for both age-based eligibility for mandated coverage and the amount of coverage available to the child under the mandate.

After adjusting for predisposing, enabling, need-based, and contextual covariates along with state and year fixed effects, each $10,000 increase in mandate generosity was associated with a 1.23 percentage-point increase in the predicted probability of behavioral treatment receipt (95% CI, 0.31 to 2.16; $p = 0.009$). Mandate generosity was also positively associated with unmet health care need (AME = 0.85 percentage points; 95% CI, 0.20 to 1.50; $p = 0.010$), while the association with problems paying medical bills was not statistically significant (AME = 0.65 percentage points; 95% CI, −0.19 to 1.50; $p = 0.131$).

Taken together, these results suggest that more generous and expansive coverage may facilitate behavioral treatment access for families. At the same time, generous coverage alone may not sufficiently alleviate broader access barriers or financial strain. The significant positive association with unmet need is consistent with two interpretations. First, children with ASD often have unmet needs across multiple domains of care, and greater engagement with the health system may make those needs more visible to families.[16] Second, rising demand for services may outpace the available provider capacity in areas where coverage is more generous, leaving families with recognized but unfulfilled needs.[17,24] The null finding for problems paying medical bills likely reflects the fact that this NSCH measure captures overall health care financial strain rather than ASD-specific spending, so improvements in ASD coverage may be offset by costs in other areas of care.

Key Limitations

A fundamental limitation of this study is the observational, cross-sectional nature of the data. Although the analysis pools eight annual NSCH cycles, each survey year captures a distinct sample of children at a single point in time rather than following the same children longitudinally. It is therefore not possible to observe within-child changes in treatment before and after a policy shift.

A second concern involves exposure misclassification arising from the structure of U.S. insurance regulation. State benefit mandates generally apply to only state-regulated, fully insured commercial plans and do not extend to self-funded employer-sponsored plans governed by ERISA.[43] Because the NSCH identifies whether a child has private insurance but does not indicate plan type, some children assigned a positive effective cap may belong to self-funded plans exempt from the mandate, while other children coded as $0 may receive ASD coverage through voluntary employer provisions. This type of misclassification would be expected to bias associations toward the null hypothesis.

Third, the primary outcome is a dichotomous, caregiver-reported measure of any behavioral treatment in the past 12 months. It does not capture treatment modality (for example, ABA versus other behavioral approaches), intensity, frequency, duration, setting, or quality. Because annual dollar caps most directly constrain the total volume and continuity of services, our dichotomous indicator of any-versus-no treatment may also understate the relationship between mandate cap generosity and the amount of treatment families actually receive.

Fourth, there are additional sources of measurement error of the primary outcome variable. Caregiver understanding of what counts as behavioral treatment may also vary across respondents, introducing measurement error. Additionally, children may access behavioral care through schools or early intervention systems that are financed outside of private insurance, further complicating the interpretation of the NSCH measure.

Fifth, the mandate exposure and the NSCH outcome are not perfectly aligned in scope. The effective cap was derived from mandate provisions that often focus on ABA-related coverage, whereas the NSCH behavioral treatment item captures a broader caregiver-reported category that may include ABA and other behavioral approaches. If the outcome includes services that are not directly governed by the dollar caps used to construct the exposure, then the exposure is a less precise measure of the coverage relevant to the reported treatment. This measurement mismatch may attenuate the observed association.

Finally, implementation constraints may weaken the link between what a mandate covers on paper and what families can actually access. For example, prior authorization requirements and claim denial patterns may shape the availability of behavioral treatment. Residual confounding from unmeasured time-varying state-level factors correlated with implementation constraints may also remain.

Data and Measurement Opportunities

An impactful improvement for future research would be to employ data sources whose structure aligns more directly with how dollar caps operate. Commercial insurance claims data can provide granular measures of treatment hours, visit counts, spending, and out-of-pocket costs, all of which are outcomes that annual caps are designed to influence more directly than a binary indicator of any treatment. Claims data can also distinguish plan funding type (fully insured versus self-funded), which would reduce the exposure misclassification that limits our analyses.

Future investigations should also incorporate more precise measures of the ASD-specific workforce to test whether policies that enhance the generosity of coverage influence service use through changes in the workforce that deliver behavioral therapies. More specifically, behavioral treatment access depends on a specialized provider supply, particularly board-certified behavior analysts.[24] State-year measures of BCBA supply, geographic distribution, and trends in certification could be integrated into models to test whether workforce availability mediates the association between coverage generosity and treatment utilization.

Linking insurance utilization records with education-sector data may also strengthen interpretation. Many children with ASD receive behavioral supports through school systems or IDEA Part C early intervention programs, which may be financed outside private insurance or through mixed public funding streams.[48,49] If mandates shift behavioral treatment onto private coverage without changing the total amount of services a child receives, or conversely if mandates expand total services, single-system analyses may miss these effects.

Methodological Extensions

Direct investigation of mechanisms is important. Future studies should test whether mandate generosity operates through identifiable pathways, such as increased provider supply, greater network participation, or lower out-of-pocket costs, by incorporating contextual state-level measures and plan-level features into analytic models. Claims or electronic health record data that distinguish ABA from other modalities and capture treatment hours, visit frequency, and duration would allow researchers to determine whether more generous mandates produce meaningful differences in treatment intensity and continuity, not just in the probability of receiving any treatment at all.

Qualitative and Mixed-Methods Opportunities

Quantitative analyses of mandate effects can identify patterns in aggregate data, but they may not explain why or how families experience them. Mixed-methods designs may address this gap; for instance, analyses could begin with quantitative models that identify policy environments or where generosity appears to matter more or less and then use qualitative interviews to uncover the mechanisms and barriers that account for this variation. Family interviews can shed light on the practical difficulties of navigating authorization requirements, locating available in-network providers, managing waitlists, and coordinating care across multiple systems. Provider and payer interviews can illuminate the factors that influence network participation decisions, the adequacy of reimbursement rates for behavioral services, and the administrative processes that may impede or facilitate timely access to treatment.

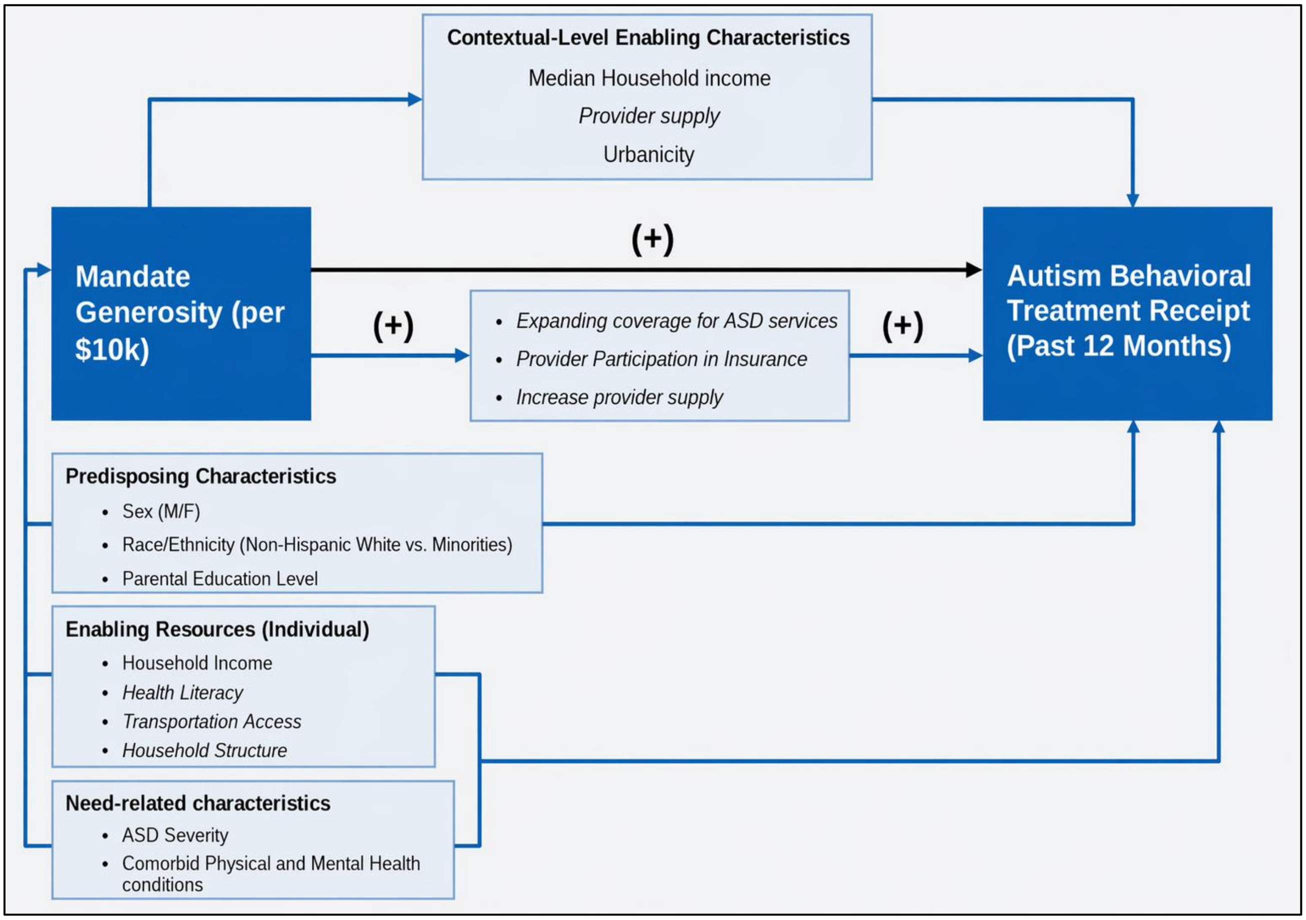
Family-centered qualitative research could also clarify what caregivers mean when they report "behavioral treatment" in surveys, how treatment differs by the child's age and context, and what types of care and support families find most valuable. These insights can inform the construction of more robust survey measures, guide the selection of claims codes that more precisely capture interventions of interest, and ultimately contribute to policy designs that reflect the actual experiences and priorities of families navigating the ASD service landscape.

Conclusion

The limitations and future directions outlined in this chapter highlight important implications and next steps for autism policy research. Richer data, complementary analytic approaches, and more direct evidence on family experiences could clarify when more generous mandates translate into meaningful gains in treatment access and when they do not. Together, those efforts can help inform policy strategies that pair broader coverage with workforce development, network adequacy, and other system-level supports so that statutory coverage is more likely to translate into timely, accessible care for children with ASD.

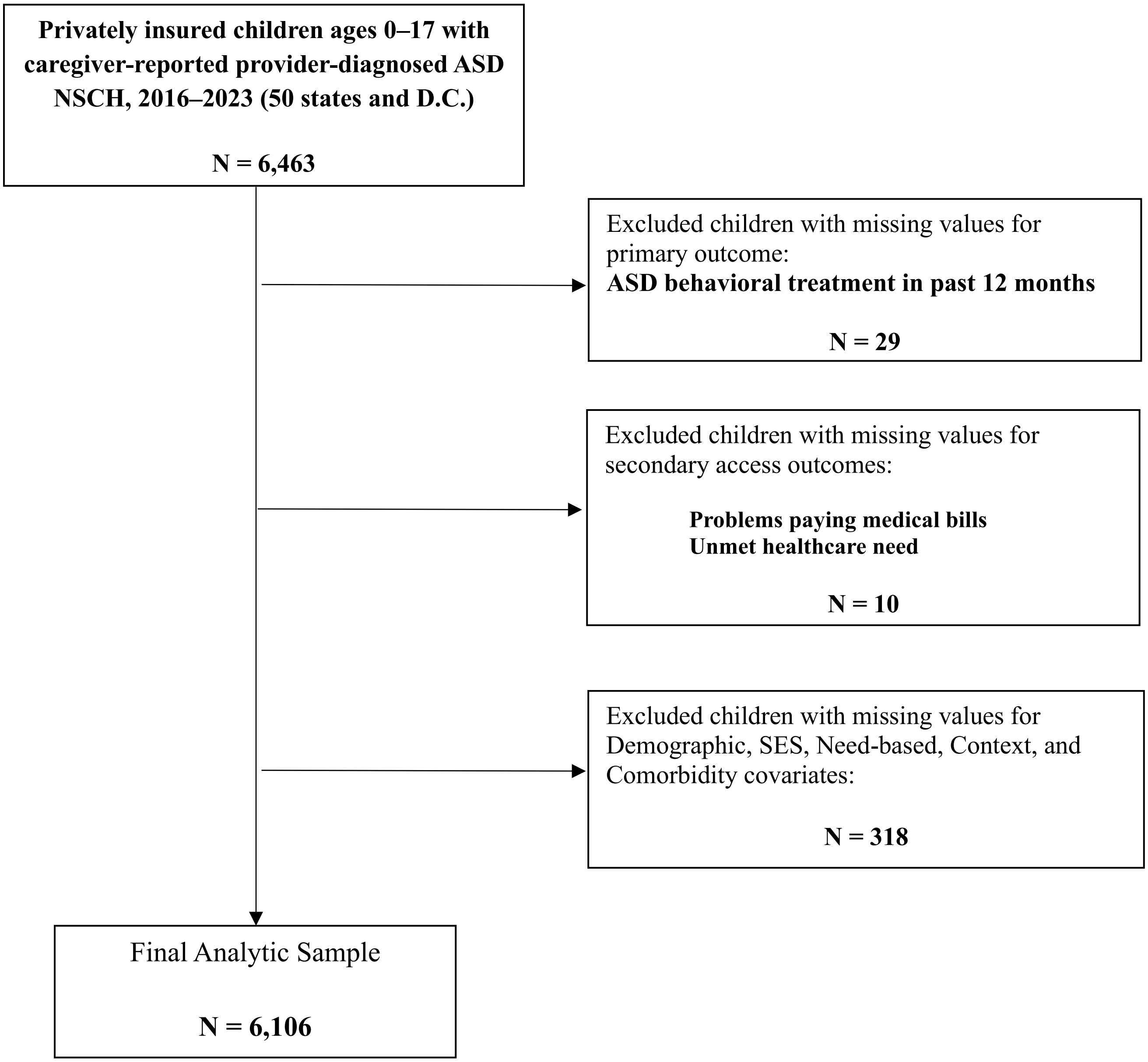
Exhibits and Appendix

Exhibit 1. Conceptual framework



Notes: Signs indicate hypothesized directional relationships. Italicized elements represent constructs that are conceived to shape the focal relationship but are not directly measured in the NSCH.

Exhibit 2. Analytic sample derivation for the ASD behavioral treatment analytic cohort



Notes: Counts reflect the unweighted analytic sample. Observations with missings for Urbanicity and ASD severity were retained using an explicit "Missing" category where applicable (no additional exclusions for missingness). Because problems paying medical bills is asked conditionally and has item nonresponse in the NSCH, the model-specific sample size for that outcome is smaller (N=5,435) than the primary analytic cohort (N=6,106).

Exhibit 3. Sample characteristics

Characteristic	Category	Unweighted n	Weighted %
Predisposing			
Child sex	Male	4,745	77.9
	Female	1,361	22.1
Race/ethnicity	NH White	4,263	56.7
	NH Black	319	11.6
	Hispanic	706	21.3
	Asian/Other (combined)	818	10.4
Enabling			
Parent education	≤ High school	497	14.6
	Some college/Assoc	1,294	19.7
	≥ Bachelor's	4,315	65.8
Household income (FPL)	<100% FPL	324	6.3
	100–199%	769	18.4
	200–399%	2,139	36.3
	≥400%	2,874	38.9
Urbanicity	Non-metro	636	8.1
	Metro	4,205	81.2
	Missing	1,265	10.7
Need			
ASD severity	Mild	3,114	48.5
	Moderate	2,089	35.7
	Severe	496	8.2
	Missing	407	7.7
Physical health comorbidities	0 (None)	4,165	69.4

Characteristic	Category	Unweighted n	Weighted %
	1 condition	1,619	25.5
	2+ conditions	322	5.1
Mental health comorbidities			
Anxiety	No	3,305	57.5
	Yes	2,801	42.5
Depression	No	5,030	84.2
	Yes	1,076	15.8
Neurodevelopmental comorbidities			
Developmental delay	No	2,103	33.4
	Yes	4,003	66.6
ADD/ADHD	No	3,364	56.5
	Yes	2,742	43.5
Other Neurodevelopmental comorbidities	0 (None)	4,968	80.3
	1 condition	920	16.0
	2+ conditions	218	3.7

Notes: Unweighted Ns. Percentages are survey-weighted using NSCH weights and survey design.

Exhibit 4. Bivariate associations between effective mandate generosity and study outcomes

Outcome	Outcome category	Unweighted n	Weighted %	Mean policy generosity, $10,000 units (SD)	Reference	*p-value*
ASD behavioral treatment	No	2,503	42.9	6.48 (3.56)	Ref	
	Yes	3,603	57.1	6.71 (3.46)		0.205
Problems paying medical bills	No	4,060	71.1	6.73 (3.54)	Ref	
	Yes	1,375	28.9	6.35 (3.43)		0.071
Unmet health care need	No	5,425	88.3	6.61 (3.52)	Ref	
	Yes	681	11.7	6.63 (3.45)		0.956

Notes: Unweighted Ns. Weighted percentages account for NSCH survey weights and complex survey design. Pairwise p-values are design-based F tests from survey-adjusted linear regression comparing outcome with "No" as the reference

Exhibit 5. Unadjusted average marginal effects from survey-weighted logistic regression models

Predictor	**ASD behavioral treatment (past 12 months) N=6,106**		**Problems paying medical bills (past 12 months) N=5,435**		**Unmet health care need (past 12 months) N=6,106**	
	AME [95% CI]	***p-value***	**AME [95% CI]**	***p-value***	**AME [95% CI]**	***p-value***
Mandate generosity, per $10,000 increase	+0.45 [-0.25, 1.15]	0.204	-0.62 [-1.29, 0.05]	0.071	+0.01 [-0.45, 0.47]	0.956

Notes: Estimates from unadjusted survey-weighted logistic regression models fit separately for each outcome. Unadjusted models include policy exposure only. Entries are average marginal effects (AME) with 95% confidence intervals in brackets. AMEs are reported as percentage-point changes in predicted probability

Exhibit 6. Adjusted average marginal effects from Survey-Weighted Two-way Fixed Effects Logistic Regression Models

Predictor	ASD behavioral treatment (past 12 months) N=6,106		Problems paying medical bills (past 12 months) N=5,435		Unmet health care need (past 12 months) N=6,106	
	AME [95% CI]	*p-value*	AME [95% CI]	*p-value*	AME [95% CI]	*p-value*
Mandate generosity, per $10,000 increase	+1.23 [0.31, 2.16]	0.009	+0.65 [-0.19, 1.50]	0.131	+0.85 [0.20, 1.50]	0.010
Sex (ref: Male)						
Female	-4.31 [-9.78, 1.15]	0.122	-0.60 [-5.80, 4.60]	0.821	-0.94 [-4.14, 2.26]	0.565
Race (ref: NH White)						
NH Black	-6.99 [-15.82, 1.83]	0.120	+8.12 [-0.54, 16.77]	0.066	-2.79 [-7.69, 2.12]	0.265
Hispanic	-6.84 [-13.65, -0.03]	0.049	+1.77 [-5.02, 8.56]	0.610	+2.80 [-1.80, 7.41]	0.233
Asian/Other (combined)	-0.14 [-6.19, 5.90]	0.963	-5.84 [-11.66, -0.02]	0.049	+0.63 [-3.58, 4.84]	0.769
Parent education (ref: ≤High school)						

Some college/Associate	+4.86 [-4.24, 13.96]	0.295	-1.65 [-11.43, 8.12]	0.740	+1.38 [-3.84, 6.59]	0.605
≥Bachelor's degree	+6.35 [-2.19, 14.90]	0.145	-7.91 [-16.56, 0.74]	0.073	+1.23 [-3.62, 6.08]	0.620
Household income (%FPL) (ref: <100%)						
100–199%	+6.57 [-3.98, 17.13]	0.222	-8.91 [-20.04, 2.21]	0.116	-0.92 [-7.46, 5.63]	0.784
200–399%	+3.79 [-5.82, 13.39]	0.440	-7.51 [-17.58, 2.55]	0.143	-0.60 [-6.70, 5.50]	0.847
≥400%	+12.34 [2.83, 21.85]	0.011	-18.47 [-28.47, -8.48]	<0.001	-6.26 [-12.16, -0.35]	0.038
Urbanicity (ref: Non-metro)						
Metro	+9.41 [3.24, 15.59]	0.003	-0.74 [-7.51, 6.02]	0.829	+3.11 [-0.07, 6.29]	0.056
Unknown	+15.84 [5.05, 26.63]	0.004	-0.66 [-12.08, 10.75]	0.910	+15.34 [5.26, 25.41]	0.003
ASD severity (ref: Mild)						
Moderate	+14.89 [9.15, 20.63]	<0.001	+5.51 [-0.07, 11.10]	0.053	+0.85 [-2.57, 4.28]	0.625
Severe	+32.44 [26.15, 38.73]	<0.001	+9.14 [0.75, 17.53]	0.033	+9.62 [3.03, 16.22]	0.004

Unknown	-31.54 [-38.64, -24.44]	<0.001	-4.20 [-11.77, 3.37]	0.277	-1.10 [-6.82, 4.62]	0.706
Physical comorbidities (ref: 0 conditions)						
1 condition	-2.14 [-7.11, 2.84]	0.400	+6.52 [1.17, 11.86]	0.017	+3.59 [0.09, 7.10]	0.045
2+ conditions	-9.54 [-19.15, 0.07]	0.052	+21.60 [12.25, 30.96]	<0.001	+11.78 [4.75, 18.82]	0.001
Mental Health comorbidities						
Anxiety (ref: No): Yes	+7.24 [1.88, 12.61]	0.008	+7.96 [2.92, 13.01]	0.002	+6.86 [2.99, 10.74]	0.001
Depression (ref: No): Yes	+1.54 [-6.04, 9.12]	0.691	+0.37 [-5.75, 6.49]	0.906	+4.90 [0.74, 9.06]	0.021
Neurodevelopmental comorbidities						
Developmental delay (ref: No): Yes	+12.48 [7.17, 17.79]	<0.001	+2.17 [-3.02, 7.36]	0.413	-2.49 [-6.07, 1.08]	0.171
ADD/ADHD (ref: No): Yes	+2.60 [-2.09, 7.29]	0.277	+5.10 [0.35, 9.85]	0.035	+4.54 [1.20, 7.88]	0.008
Other Neurodevelopmental comorbidities (ref: 0 conditions)						
1 condition	-3.75	0.261	+0.81	0.800	+4.57	0.038

	[-10.28, 2.78]		[-5.43, 7.04]		[0.26, 8.89]	
2+ conditions	-7.09 [-20.47, 6.28]	0.298	-0.39 [-12.00, 11.23]	0.948	+3.66 [-3.68, 10.99]	0.328
State-level covariates						
Median household income (per $10,000)	+2.66 [-4.78, 10.10]	0.484	-3.92 [-11.70, 3.82]	0.321	-6.46 [-11.70, -1.26]	0.015
Psychiatrists per 100,000 (per 10-unit increase)	+2.02 [-1.48, 5.51]	0.258	-0.79 [-4.19, 2.61]	0.649	-0.90 [-2.96, 1.16]	0.392

Notes: Survey-weighted logistic regression. AMEs are reported as percentage-point changes in predicted probability. For categorical predictors, AMEs represent discrete changes from the reference category. All models include state and survey-year fixed effects (dummies omitted from table).

Exhibit 7A. Sensitivity analysis using 3-category policy exposure

Predictor	**ASD behavioral treatment (past 12 months) N=6,106**		**Problems paying medical bills (past 12 months) N=5,435**		**Unmet health care need (past 12 months) N=6,106**	
	AME [95% CI]	***p-value***	**AME [95% CI]**	***p-value***	**AME [95% CI]**	***p-value***
Mandate generosity, per $10,000 increase						
>$0 and <$36,000	+5.96 [-4.48, 16.39]	0.263	+2.23 [-6.74, 11.21]	0.626	0.00 [-3.96, 4.04]	0.985
≥$36,000	+15.34 [6.46, 24.22]	0.001	+4.78 [-2.64, 12.20]	0.207	+6.64 [2.56, 10.70]	0.001

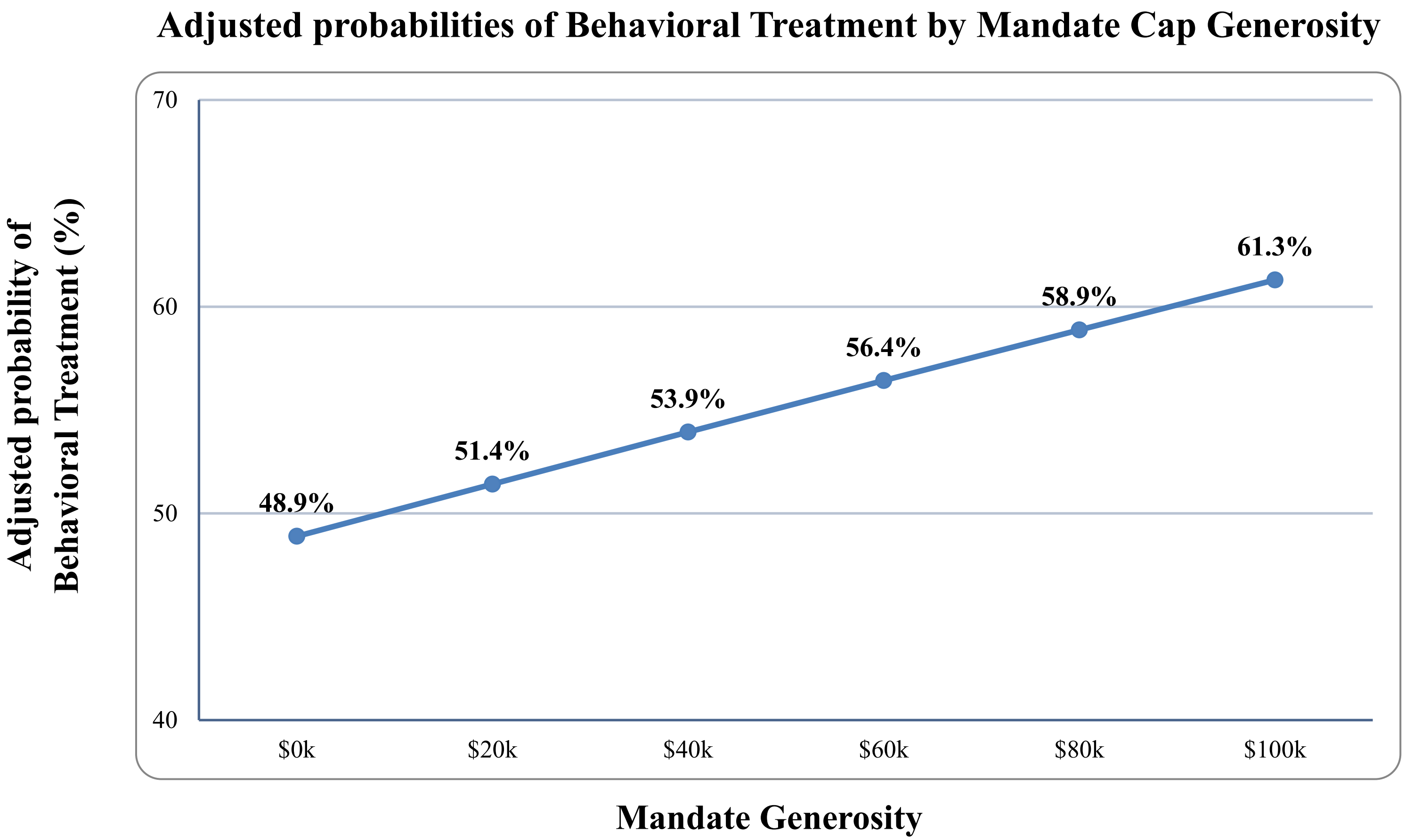
Notes: Models are survey-weighted logistic regressions with state and year fixed effects. Entries are average marginal effects with 95% confidence intervals in brackets. AMEs are reported as percentage-point changes in predicted probability

Exhibit 7B. Sensitivity analysis using binary policy exposure

Predictor	**ASD behavioral treatment (past 12 months) N=6,106**		**Problems paying medical bills (past 12 months) N=5,435**		**Unmet health care need (past 12 months) N=6,106**	
	AME [95% CI]	***p-value***	**AME [95% CI]**	***p-value***	**AME [95% CI]**	***p-value***
Mandate generosity, per $10,000 increase						
Any positive effective cap (> $0)	+11.78 [3.34, 20.22]	0.006	+3.96 [-3.13, 11.04]	0.274	+4.63 [0.81, 8.45]	0.018

Notes: Models are survey-weighted logistic regressions with state and year fixed effects. Entries are average marginal effects with 95% confidence intervals in brackets. AMEs are reported as percentage-point changes in predicted probability

Exhibit 8. Adjusted probability of Behavioral Treatment by Mandate Cap Generosity



Notes: Points represent adjusted predicted probabilities of ASD behavioral treatment receipt from the fully adjusted survey-weighted logistic regression model. Predictions were generated across values of the effective annual ASD coverage cap ($0 to $100,000).

Appendix Table A1. Need-related comorbidity measures and conditions

Measure in analysis	Conditions included / description	Coding used in models
Other neurodevelopmental comorbidity count	Deafness/hearing problems; epilepsy/seizure disorder; Down syndrome; intellectual disability. Developmental delay and ADD/ADHD were not included in this count because they were modeled separately.	0 conditions; 1 condition; 2+ conditions
Physical comorbidity count	Asthma; heart condition; diabetes; gastrointestinal/stomach condition	0 conditions; 1 condition; 2+ conditions
Developmental delay	Modeled separately as its own binary covariate; not included in the other neurodevelopmental comorbidity count	No; Yes
ADD/ADHD	Modeled separately as its own binary covariate; not included in the other neurodevelopmental comorbidity count	No; Yes
Anxiety	Modeled separately as its own binary covariate	No; Yes
Depression	Modeled separately as its own binary covariate	No; Yes
ASD severity	Caregiver-reported ASD severity	Mild; Moderate; Severe; Missing

Notes: “Other neurodevelopmental comorbidity count” excludes developmental delay and ADD/ADHD, which were modeled separately as individual binary indicators due to sample size.

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